Discriminative Spectral–Spatial Feature Extraction-Based Band Selection for Hyperspectral Image Classification

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***Abstract*— Recently, some spectral–spatial band selection (BS) strategies have become hugely popular as they fuse the spectral information of the pixels and the spatial relationship with the neighboring pixels to enhance the performance of classification methods. However, being unsupervised in nature, these methods do not utilize the class information of the training samples which could substantially empower the capabilities of such spectral–spatial BS methods. To circumvent this limitation, a supervised spectral–spatial BS method based on component loadings obtained from the principal components of spectral–spatial principal component analysis (PCA) and using a novel super-pixel based graph Laplacian embedding is proposed. The methodology attempts to unify the two strategies of dimensionality reduction, i.e., BS and feature extraction (FE), so that the benefits from both of them can be combined. The importance of each band is estimated in terms of its component loadings along the principal components which are estimated from a unified objective function consisting of three terms: data fidelity, classification error term, and spatial prior. Additionally, the spatial relationship among the neighboring samples is characterized using a novel superpixel-based graph model. An objective comparison of the proposed approach with several widely used, state-of-the-art BS methods demonstrates a significant improvement in the classification accuracy.**

***Index Terms*— Band selection (BS), component loadings, hyperspectral, neighborhood, principal component analysis (PCA), simple linear iterative clustering (SLIC), spatial information, superpixel.**

I. Introduction

HYPERSPECTRAL sensors, with their advanced sensing technology, can determine the spectral reflectance of materials using hundreds of extremely narrow and contiguous bands [1]. Such detailed spectral information can help in capturing the subtle differences among materials of a similar reflectance profile [2]–[5]. However, many of these bands are spectrally correlated and thus, redundant in nature. Furthermore, processing such a large number of bands can lead to heavy computational complexity and memory consumption. Additionally, more samples are required to model and approximate the probability distribution function of the hyperspectral data [6], [7]. An insufficient number of samples can be distressed by overfitting and reduced generalization capability of classifiers [8], [9]. Thus, there arises a need for dimensionality reduction of the hyperspectral data to address the above-mentioned challenges faced in processing high-dimensional data.

Dimensionality reduction of hyperspectral data can be performed under two paradigms [10], [11]: feature extraction (FE) and band selection (BS). FE methods such as principal component analysis (PCA), linear discriminant analysis (LDA), Fourier transform, etc. transform the original data, linearly or nonlinearly, into another feature space. Some of the supervised methods such as LDA, spatial–spectral hypergraph discriminant analysis (SSHGDA) [12], etc. aim to improve the discriminative capability of the features for better classification. For instance, SSHGDA uses between-class scatter matrix, weighted within-class scatter matrix, intraclass spatial–spectral hypergraph, and interclass spatial-spectral hypergraph to reveal the complex spatial–spectral structures of hyperspectral image and enhance the interclass separability. Some FE methods also aim to bring out the intrinsic spatial information of the image which can enhance the performance of classification methods. For instance, the semisupervised sparse manifold discriminative analysis (S3MDA) method in [13] uses the manifold-based sparse representation (MSR) and graph embedding (GE) to enhance interclass separability and intraclass compactness for better discrimination between classes. BS, on the other hand, selects the desired band subset from the original set of bands. However, FE methods alter the original values of reflectances which can complicate their interpretation necessary for various applications and analysis. BS methods, on the other hand, keep the spectral meaning of the original bands preserved [14], [15] which results in better interpretation of the behavior of the materials. This has necessitated the need for FE-based BS methods to capture the valuable information from the extracted features but at the same time, to ensure the interpretability of the material properties in terms of their original reflectance values. Several noted approaches based on PCA [14], scale-invariant feature transform (SIFT) [16], independent component analysis (ICA) [17], LDA [18], or wavelet

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transform [19] have been attempted to bridge the two dimensionality reduction paradigms.

BS methods can be categorized as supervised and unsupervised. Unsupervised BS methods measure the significance of the bands based on statistical characteristics like non-Gaussianity, correlation, similarity [10], etc. while the supervised methods use labeled training samples to select discriminative bands based on criteria like class separability, divergence, etc. The supervised approach requires labeled training data which is difficult to acquire but gives better performance in comparison to the unsupervised methods. Some well-known supervised BS methods are minimum misclassification canonical analysis (MMCA) [18], minimum estimated abundance covariance (MEAC) [20], spectral angle mapper-based band add-on (SAM-BAO) [21], feature weighting BS [22], phase correlation [23], etc.

Recently, spectral–spatial BS methods have gained huge popularity as they exploit the spatial relationship among the neighboring pixels to enhance the performance of classification methods. These methods can control the salt and pepper appearance of the classification map which occurs due to several factors such as atmospheric or instrumental noise. Some well-known methods include saliency bands and scale selection (SBSS) [24], manifold-preserving and weakly redundant (MPWR) [25], efficient graph convolutional self-representation (EGCSR) [26], etc. These methods are unsupervised in nature and do not exploit the available labeled training data which could substantially empower the capabilities of the spectral–spatial BS methods.

In this article, a supervised spectral–spatial BS method based on component loadings obtained from the principal components of spectral–spatial PCA and using a novel superpixel-based graph Laplacian embedding is proposed. In this method, the traditional objective function of PCA is modified by incorporating three terms: data fidelity, classification error (which incorporates the class information from the labeled training samples), and the spatial prior (which characterizes the spatial relationship among neighboring pixels). The spatial relationship among the samples is characterized using a novel superpixel-based graph model, where the superpixels were created using simple linear iterative clustering (SLIC) [27] algorithm. The importance of each band is represented in terms of its component loadings [14], [28]–[30] with the top principal components. The complete operation is applied in a cluster-wise fashion by first partitioning the complete set of bands into highly correlated subgroups and applying the discriminative, spectral–spatial PCA on each cluster to select band(s) representing each cluster. The main contributions of the proposed methodology are summarized as follows.

1) The bands are selected using the component loadings along the principal components which are estimated from a unified objective function consisting of three terms: data fidelity, classification error term, and spatial prior.
2) The spatial term in the objective function of PCA is incorporated using a novel superpixel-based graph model which minimizes search space and time in comparison to the traditional window-based approach.
3) Extensive experiment studies demonstrate that the proposed algorithm is able to outperform the latest, state-of-the-art approaches for a smaller training size and lesser number of bands.

The remainder of the article is organized as follows: Section II-A gives an overview of the different supervised and spectral–spatial BS methods. Section II-B reviews the PCA-based dimensionality reduction methods. Section III presents the proposed loadings-based BS strategy using discriminative spectral–spatial PCA. In Section IV, comprehensive experiments are conducted on three hyperspectral datasets to compare the method with both new and widely accepted, state-of-the-art BS paradigms, and the results are discussed and analyzed in detail. Finally, the conclusion of the article is given in Section V.

II. Related Work

In this section, different BS methods are discussed. This is followed by a brief overview of different PCA-based dimensionality reduction methods as it is the basic framework in the proposed framework.

A. Review of BS Methods

Supervised BS algorithms give better classification performance in comparison to unsupervised methods [10] due to the use of labeled training samples. The existing supervised BS methods can be categorized as ranking, searching, or low rank (LR) optimization-based. Ranking-based methods use indices and measures based on criteria such as Fisher score [31], mutual information [32], PCA [22], and phase correlation [23] to establish the relationship between the bands and reference/label map and accordingly, rank the bands on the basis of their discriminative quality. For instance, MMCA [18] uses classification-based band prioritization criterion to rank and select bands. However, in ranking-based methods, the selected bands may be redundant as they are used to maximize the same objective function. Searching-based methods look for optimal band combinations, either exhaustively or incrementally, to maximize separation between different classes. The methods aim to select bands that are discriminative as well as distinct. Different searching-based methods use measures such as angular separation [23], optimal index factor [33], and mutual information [34] to select bands in an iterative fashion. LR optimization-based methods use optimization problems imposing LR and/or sparsity constraints on the coefficient matrix which reveal the underlying structures within hyperspectral image (HSI) data needed for clustering band into different groups. Here, the objective functions are customized to describe the correlation [35] or dissimilarity [36] among different bands. Some other LR-based supervised BS methods include $l_{2,1}$ regularization-based least square regression model for maximization of interclass distance [37] or l_1 constrained log-likelihood function representing the correlation between bands and classification output [38].

Lately, deep neural networks (DNNs)-based BS methods have also gained huge success. For instance, in [39], a deep convolutional neural network (CNN) and rectified linear unit (ReLU)-based BS method tests the different band

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combinations without retraining the model each time, using a new subspace-partitioning method and distance density (DD). An unsupervised BS method for target detection based on deep latent spectral representation learning (DLSRL) [40], searches the closest representation of the original HSIs by imposing multivariate Gaussian, spectral consistency, and physical significance on deep latent space. In reinforcement learning for semi supervised-based band selection (RLSBS) [41], the BS is formalized as a reinforcement learning problem. Here, the bands are selected using a semisupervised CNN, called EvaluateNet which uses both limited labeled and sufficient unlabeled samples under the intraclass compactness constraint. A novel CNN method based on band-wise independent convolution and hard thresholding (BHCNN) [42] combines BS, multiscale spatial–spectral FE, and classification into a unified optimization procedure.

Several factors such as atmospheric effects, natural spectrum variation, instrument noise, and incident illumination lead to spectral variability within a class and erroneous labeling of the samples. Noiseless, smooth classification of high-resolution, complex scenes requires a combination of both spectral and spatial information from the hyperspectral dataset. For instance, pixels that are spatially adjacent to each other share similar spectral characteristics. By combining the spectral information of these spatially close pixels, misclassification of anomalous or spectrally uncertain samples can be controlled, resulting in smoother, and less noisier classification results. For instance, in [24], rare, noise-free, saliency bands which are capable of characterizing the homogenous structures and objects in the image are selected using the trace and determinant of the Hessian matrix. An unsupervised BS called MPWR [25] selects uncorrelated bands which are rich in structural information and can retain the manifold structure of the dataset. The hypergraph-based band scoring method in [39] selects those bands that best capture the spectral and spatial relationship between a group of pixels. EGCSR BS method [26] incorporates spectral–spatial graph convolution into the self-representation model to segment bands into different subspaces. A spectral–spatial BS framework in [43] maximizes the multiobjective cost function incorporating clustering (spectral) and intraband (spatial) decorrelation measure (Frobenius norm). In [44], a structural similarity-based ranking method is used to measure the spatial relationship between bands for the selection of cluster representatives. A hybrid-graph learning method termed enhanced hybrid-graph discriminant learning (EHGDL) [45] combines intraclass and interclass hypergraphs to reveal the complex high-order relationships among different samples. A local-view-assisted discriminative band selection method with hypergraph autolearning (LvaHAl) [46] selects discriminative bands with both local and global perspectives.

B. Brief Overview of PCA-Based Dimensionality Reduction Methods

This section gives a brief overview of different PCA-based dimensionality reduction methods as the proposed method is built upon the objective function of PCA.

PCA aims to reduce the dimensionality of a dataset consisting of a large number of correlated variables while retaining as much variance in the dataset as possible.

Assume that $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_m] \in \mathbb{R}^{d\times m}$ is a high-dimensional, hyperspectral dataset consisting of m number of pixels, d number of bands and c classes. The objective of PCA is to obtain $\mathbf{Q} = [\mathbf{q}_1, \mathbf{q}_2, \ldots, \mathbf{q}_m] \in \mathbb{R}^{k\times m}$ which is k-dimensional representation of observed d-dimensional data $\mathbf{X}$, where $k <= d$. The objective function can be written as

$$\min_{\mathbf{Y},\mathbf{Q}} \left\|\mathbf{X} - \mathbf{Y}\mathbf{Q}^{\mathbf{T}}\right\|_F^2, \quad \text{s.t. } \mathbf{Y}^{\mathrm{T}}\mathbf{Y} = \mathrm{I} \tag{1}$$

where $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_k] \in \mathbb{R}^{k\times d}$ is the projection matrix to the principal components corresponding to the top k Eigen values characterizing the maximum variance of the data. However, PCA is an unsupervised learning method; it fails to incorporate any discriminative information about the different classes. Although, some may consider LDA as a suitable supervised dimensionality reduction method, it fails when the sample size is small or the number of classes is less [48].

Several attempts [49]–[51] have been made to integrate class information in the objective function of PCA to obtain low-rank representation of the whole data. Similarly, spatial information of HSI has also been incorporated to improve the performance of PCA-based dimensionality reduction approaches. For instance, in [52], each pixel, along with its neighbors, is jointly treated as a single sample for computing the spectral–spatial principal components. In [53]–[55], PCA is combined with graph-Laplacian embedding-based regularization to learn the low-dimensional representation of the dataset that incorporates its spatial information. A clustering-based BS using graph-regularized fast and robust PCA (FRPCA) [55] is proposed to select bands with high signal-to-noise (SNR) ratio.

III. Proposed Methodology

The algorithm proposed here aims to select bands that exploit the spectral as well as spatial information to provide maximum discrimination among the classes. For this, a novel variant of PCA called discriminative, spectral–spatial principal component analysis (DSS-PCA) is proposed. This helps in effectively discerning the differences among different classes on the basis of their spectral as well as structural information. The workflow of the proposed approach is shown in Fig. 1. The BS methodology consists of two stages: In the clustering stage, bands are divided into different clusters on the basis of their spectral similarity. In the ranking stage, top-ranking bands are selected to represent each cluster. For this, DSS-PCA is applied on each cluster and component loadings of DSS-PCA are used to rank the bands in the cluster and select the most discriminative and informative bands as cluster representatives. The final set of bands consists of the selected representatives from all the clusters. The proposed superpixel-based DSS-PCA is discussed in detail in Section III-A followed by a discussion on the BS framework using loadings of DSS-PCA in Section III-B.

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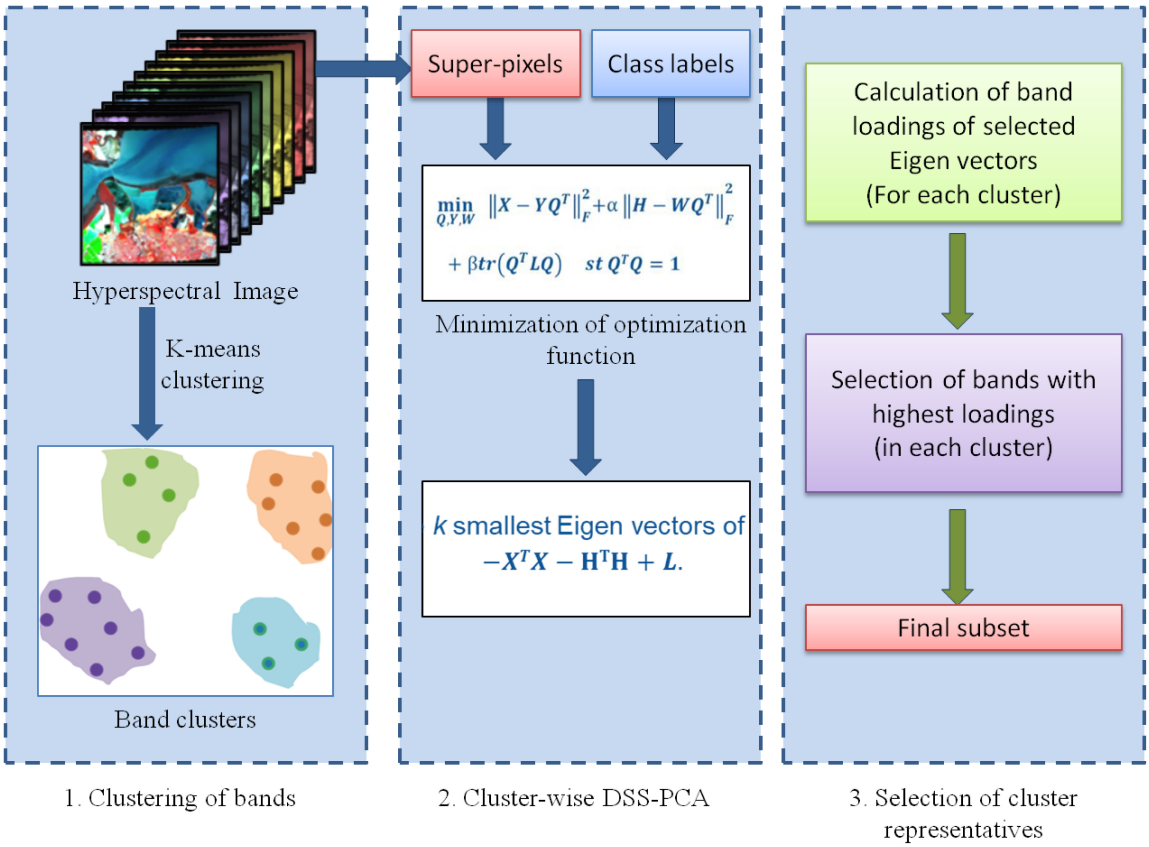


Fig. 1. Schematic representation of the proposed BS framework: The algorithm has three main stages. In first stage, the bands are grouped into P clusters using k-means clustering method. In second stage, on every cluster, the modified objective function of PCA incorporating three terms: data fidelity, classification error (using labeled training samples), and the spatial prior (which characterizes the spatial relationship among neighboring pixels) is applied and the top principal components are selected. The spatial relationship among the samples is characterized using a novel superpixel-based graph model, where the superpixels were created using SLIC algorithm. Finally, in the last stage, bands with highest loadings for the top PCs in each cluster are chosen as cluster representatives.

A. Discriminative Spectral–Spatial PCA (DSS-PCA)

The proposed method performs the modification in the traditional objective function of PCA shown in (1) by integrating two terms to it; first is the classification loss term and second is the spatial information to characterize the similarity among neighboring pixels. Following this, the unified objective function can be written as:

$$\min_{\mathbf{Y},\,\mathbf{Q},\,\mathbf{W}} \left\|\mathbf{X} - \mathbf{Y}\mathbf{Q}^{\mathbf{T}}\right\|_F^2 + \alpha \left\|\mathbf{H} - \mathbf{W}\mathbf{Q}^{\mathbf{T}}\right\|_F^2 + \beta\, \mathrm{Tr}\left(\mathbf{Q}^{\mathrm{T}}\mathbf{L}\mathbf{Q}\right) \quad \text{s.t. } \mathbf{Q}^{\mathrm{T}}\mathbf{Q} = \mathrm{I} \tag{2}$$

where α and β are the two scalar parameters and are used to control the relative contribution of the two terms. Tr(.) is the trace of the matrix.

The first term in (2) is an alternate representation of the objective function of PCA shown in (1), as defined in [51], [53], and [54], where the orthonormality is imposed on $\mathbf{Q}$. The second term in (2) is a classification error function to be minimized and it ensures that the projected data not only has representational power but also has complementary, discriminative ability. Here, $\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_m] \in \mathbb{R}^{c \times m}$ is a label matrix or class indicator for c number of classes and m number of labeled training samples [47]–[49] defined as

$$\begin{bmatrix} 0 & 0 & 0 & 1 & \cdots & 0 \\ 0 & 0 & 0 & 0 & \cdots & 1 \\ \vdots & & \ddots & & \vdots & \\ 0 & 1 & 0 & 0 & \cdots & 0 \end{bmatrix}. \tag{3}$$

This matrix comprises of 0's and 1's such that $\mathbf{h}_i = [0, \ldots, 1, \ldots, 0]^T$ is the label vector representing the class information of sample $\mathbf{x}_i$ belonging to l-th class c_l and all the entries of $\mathbf{h}_i$ are zero except l-th entry. When $\mathbf{Q}$ is fed into a linear classifier $\mathcal{F}(\mathbf{W}, t, \mathbf{Q}) = \mathbf{W}\mathbf{Q}^{\mathrm{T}} + t$ with classifier parameters $\mathbf{W}$, t, the output of the linear classifier is comparable with the label matrix $\mathbf{H}$ such that $\mathbf{H} = \mathbf{W}\mathbf{Q}^{\mathrm{T}} + t$. The objective function is minimized after setting $t = 0$ for the sake of simplicity and arbitrarily initializing $\mathbf{W}$.

The third and last term in (2) incorporates local spatial information into PCA as suggested in [53]. However, in this work, a new and improved superpixel-based variant of the traditional graph Laplacian embedding-based approach [56], [57] is proposed. It aims to minimize the following embedding cost function such that the connected components or the neighbors remain as close to each other as possible after projection along

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the PCs:

$$\min_{\mathbf{Q}} \mathrm{Tr}\left(\mathbf{Q}^{\mathrm{T}}\mathbf{L}\mathbf{Q}\right), \quad \text{s.t. } \mathbf{Q}^{\mathrm{T}}\mathbf{Q} = \mathrm{I} \tag{4}$$

where $\mathbf{L} = \mathbf{D} - \mathbf{A}$ is the Laplacian matrix obtained from the affinity matrix $\mathbf{A} \in \mathbb{R}^{m\times m}$ and the degree matrix $\mathbf{D} \in \mathbb{R}^{m\times m}$. $\mathbf{A}$ characterizes the local neighborhood relationship in terms of pair-wise similarities between a sample/pixel and its nearest neighbors which are generally selected using Euclidean distance measure. In the proposed model, it is assumed that pixels that are connected components or neighbors of each other in the graph model have a tendency to belong to the same superpixel as they are both spectrally as well as spatially closer to each other than the rest of the pixels. The approach has two advantages. First, it reduces the search space of finding the neighbors of a pixel from the global space (whole image) to only within a superpixel, which effectively saves a lot of time in computing $\mathbf{A}$. Second, pixels belonging to the same superpixel have greater probability of belonging to the same class. Thus, they are the most suitable nearest neighbors of each other. The superpixel-based graph Laplacian embedding model can be defined as follows:

$$\mathbf{A}(i,j) = \begin{cases} \exp\left(-\dfrac{\left\|\mathbf{x}_i - \mathbf{x}_j\right\|^2}{\sigma^2}\right), & \text{if } \mathbf{x}_i,\ \mathbf{x}_j \text{ belong to same superpixel} \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

where σ^2 is a user-defined parameter and $\mathbf{A}(i,j)$ is the measure of closeness between samples $\mathbf{x}_i$ and $\mathbf{x}_j$ whose value is zero if $\mathbf{x}_i$, $\mathbf{x}_j$ belong to different superpixels. Using $\mathbf{A}$, $\mathbf{D}$ and $\mathbf{L}$ are calculated. Similar pixels will possess larger values of $\mathbf{A}(i,j)$ enforcing the differences between their low-dimensional transformed representation $\mathbf{q}_i$, $\mathbf{q}_j$ to become smaller, thus minimizing the following term:

$$\min_{\mathbf{Q}} \frac{\sum_{i,j=1}^{m}\left\|\mathbf{q}_i - \mathbf{q}_j\right\|^2 \mathbf{A}(i,j)}{2}. \tag{6}$$

After some algebraic operations [57], this will be equivalent to (4) (Refer to Appendix I for proof).

In the proposed method, the superpixel map is generated using the SLIC algorithm as it gives regular, compact superpixels, has low computational head and only one tunable parameter in comparison to existing graph-based and gradient ascent-based superpixel methods [27]. In SLIC, the distance D' between two pixels, which is used for clustering the pixels into superpixels, is a weighed combination of spatial and spectral distance and is defined as

$$D' = \sqrt{\left(\frac{D_c}{\tau}\right)^2 + \left(\frac{D_s}{S}\right)^2} \tag{7}$$

where $D_s = ((r_i - r_j)^2 + (c_i - c_j)^2)^{1/2}$ is the spatial distance between the two samples $\mathbf{x}_i$ and $\mathbf{x}_j$ whose spatial coordinates (row and column positions) are (r_i, c_i) and (r_j, c_j) and $D_c = (\sum_{k=1}^{d}(x_{ik} - x_{jk})^2)^{1/2}$ is the color distance between the two samples computed using spectral reflectances of all d bands, S is the grid sampling interval defined for all pixels in the image and N_s superpixels and τ is the tunable compactness parameter in SLIC algorithm.

It should be noted that if K neighbors are used to define the relationship between $\mathbf{x}_i$, $\mathbf{x}_j$ instead of superpixel, the proposed method becomes equal to the conventional graph Laplacian Embedding in which affinity matrix is defined as follows:

$$\mathbf{A}(i,j) = \begin{cases} \exp\left(-\dfrac{\left\|\mathbf{x}_i - \mathbf{x}_j\right\|^2}{\sigma^2}\right), & \text{if } \mathbf{x}_i, \mathbf{x}_j \text{ are } k\text{-neighbors} \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

B. Optimization and Closed Form Solution

The objective function has three variables, **Y, W, Q** and its minimization problem can be simplified by use of partial derivatives. The objective function is minimized with respect to each variable by keeping the rest of the variables fixed. From (2), let

$$\mathcal{L}(\mathbf{Y},\mathbf{Q},\mathbf{W}) = \left\|\mathbf{X} - \mathbf{Y}\mathbf{Q}^{\mathbf{T}}\right\|_F^2 + \alpha\left\|\mathbf{H} - \mathbf{W}\mathbf{Q}^{\mathbf{T}}\right\|_F^2 + \beta\,\mathrm{Tr}\left(\mathbf{Q}^{\mathrm{T}}\mathbf{L}\mathbf{Q}\right). \tag{9}$$

Taking the partial derivative of $\mathcal{L}(\mathbf{Y},\mathbf{Q},\mathbf{W})$ with respect to $\mathbf{Y}$ while keeping $\mathbf{W},\mathbf{Q}$ fixed, followed by setting $(\partial\mathcal{L}/\partial\mathbf{Y}) = 0$ gives

$$\frac{\partial\mathcal{L}}{\partial\mathbf{Y}} = -2\left(\mathbf{X} - \mathbf{Y}\mathbf{Q}^{\mathrm{T}}\right)\mathbf{Q} = 0 \implies \mathbf{Y} = \mathbf{X}\mathbf{Q}. \tag{10}$$

Similarly, taking the partial derivative of $\mathcal{L}(\mathbf{Y},\mathbf{Q},\mathbf{W})$ with respect to $\mathbf{W}$ while keeping $\mathbf{Y},\mathbf{Q}$ fixed will give

$$\frac{\partial\mathcal{L}}{\partial\mathbf{W}} = -2\left(\mathbf{H} - \mathbf{W}\mathbf{Q}^{\mathrm{T}}\right)\mathbf{Q} = 0 \implies \mathbf{W} = \mathbf{H}\mathbf{Q}. \tag{11}$$

Setting the values of $\mathbf{Y}$ and $\mathbf{W}$ obtained from (10) to (11), in (9) results in

$$\mathcal{L}_1(\mathbf{Q}) = \left\|\mathbf{X} - \mathbf{X}\mathbf{Q}\mathbf{Q}^{\mathbf{T}}\right\|_F^2 + \alpha\left\|\mathbf{H} - \mathbf{H}\mathbf{Q}\mathbf{Q}^{\mathbf{T}}\right\|_F^2 + \beta\,\mathrm{Tr}\left(\mathbf{Q}^{\mathrm{T}}\mathbf{L}\mathbf{Q}\right). \tag{12}$$

After some algebraic operations, this becomes equal to $\mathcal{L}_1(\mathbf{Q}) = \mathrm{Tr}(\mathbf{Q}^{\mathrm{T}}(-\mathbf{X}^{\mathrm{T}}\mathbf{X} - \alpha\mathbf{H}^{\mathrm{T}}\mathbf{H} + \beta\mathbf{L})\mathbf{Q}) + \mathrm{Tr}(\mathbf{X}^{\mathrm{T}}\mathbf{X}) + \mathrm{Tr}(\mathbf{H}^{\mathrm{T}}\mathbf{H})$ and the minimization problem becomes

$$\min_{\mathbf{Q}} \mathrm{Tr}\left(\mathbf{Q}^{\mathrm{T}}\left(-\mathbf{X}^{\mathrm{T}}\mathbf{X} - \alpha\mathbf{H}^{\mathrm{T}}\mathbf{H} + \beta\mathbf{L}\right)\mathbf{Q}\right), \quad \text{s.t. } \mathbf{Q}^{\mathrm{T}}\mathbf{Q} = \mathrm{I}. \tag{13}$$

The proof of (13) is given in Appendix II. The solution of minimization of (13), $\mathbf{Q} = \{\mathbf{q}_1, \mathbf{q}_2, \ldots, \mathbf{q}_m\} \in \mathbb{R}^{k\times m}$ is the Eigen vectors corresponding to smallest k Eigen values of $(-\mathbf{X}^{\mathrm{T}}\mathbf{X} - \alpha\mathbf{H}^{\mathrm{T}}\mathbf{H} + \beta\mathbf{L})$ or the largest k Eigen values of $(\mathbf{X}^{\mathrm{T}}\mathbf{X} + \alpha\mathbf{H}^{\mathrm{T}}\mathbf{H} - \beta\mathbf{L})$.

C. BS Framework Using Discriminative Spectral–Spatial PCA

In the proposed work, the bands are selected based on the component loadings along the PCs obtained using the methodology explained in the previous section. To ensure that bands from different wavelength ranges are selected, similar bands are first grouped together into clusters and representative

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bands from each cluster are selected using the proposed DSS-PCA approach. The final set of selected bands consists of the selected representatives from all clusters. The workflow of the proposed BS framework is illustrated in Fig. 1.

1) Clustering of Bands: Let $\mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_p$ be the different clusters/ subgroups of spectrally similar bands. The number of bands in each subgroup is denoted as $d_1, d_2, \ldots, d_p$. In the proposed method, k-means clustering was used to divide the bands into P clusters. The number of subgroups P is chosen using a cluster evaluation methods called David Bouldin measure [58].

2) Computation of Cluster-Wise DSS-PCA: Let $\mathbf{X}_1 \in \mathbb{R}^{d_1 \times m}$, $\mathbf{X}_2 \in \mathbb{R}^{d_2 \times m}, \ldots, \mathbf{X}_P \in \mathbb{R}^{d_P \times m}$ be the image subsets made of $d_1, d_2, \ldots, d_P$ bands from cluster $\mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_p$.

Using the superpixel segmentation map and (5), the weight matrices $\mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_P \in \mathbb{R}^{m \times m}$ for $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_P$ are computed. Then, using (13), the Eigen vectors of largest k Eigen values $(\mathbf{X}_i{}^{\mathrm{T}}\mathbf{X}_i + \alpha\mathbf{H}^{\mathrm{T}}\mathbf{H} - \beta\mathbf{L}_i)$ for every cluster $\mathcal{S}_i$ are computed. It should be carefully noted that as per (1) and (2), the Eigen vectors $\mathbf{Q}_i \in \mathbb{R}^{k \times m}$ are actually the normalized PCs or the projected data. As a consequence, $\mathbf{Y}_i \in \mathbb{R}^{k \times d_i}$ represents projection matrix or the principal directions and can be computed from $\mathbf{Q}_i$ using (10). $\mathbf{Y}_i$ will be used for selecting the cluster representatives as discussed next.

3) Component Loadings-Based Selection of Cluster Representatives: To select the representatives of cluster $\mathcal{S}_i = \{b_1^i, b_2^i, \ldots, b_{d_i}^i\}$, the component loadings [14], [28]–[30] of all bands within the cluster as calculated and the bands having the highest loadings with the top k components are selected as the representative bands. For cluster $\mathcal{S}_i$, the variance $\mathbf{V}_i = \{v_1^i, v_2^i, \ldots, v_k^i\}$ associated with the principal component $\mathbf{Q}_i \in \mathbb{R}^{k \times m}$ in terms of projection matrix $\mathbf{Y}_i = \{\mathbf{y}_1^i, \mathbf{y}_2^i, \ldots, \mathbf{y}_k^i\} \in \mathbb{R}^{k \times d_i}$ and $\mathbf{Q}_i$ can be defined as

$$\mathbf{V}_i = \mathbf{Y}_i{}^{\mathrm{T}}\mathbf{Q}_i\mathbf{Y}_i. \tag{14}$$

The component loadings $c_{t,k}^i$ of a band b_t^i with k-th principal component can be defined as

$$c_{t,k}^i = y_{k,t}^i \sqrt{v_k^i} \tag{15}$$

where $y_{k,t}^i$ is the coefficient for b_t^i band in the k-th component and v_k^i is the corresponding variance of the component. The combined correlation of band b_t^i with the top k principal components is given as

$$c_t^i = \sqrt{\sum_{j=1}^{k} \left(c_{t,j}^i\right)^2}. \tag{16}$$

The band with the highest value of c_t^i is chosen to represent the cluster. The complete algorithm is again summarized in the form of pseudocode in Algorithm 1.

Algorithm 1 Band Selection Using Component Loadings-Based Discriminative Spectral-Spatial PCA

Input: Hyperspectral data subset $\mathbf{X} \in \mathbb{R}^{d \times m}$ with d bands and m labeled training samples ; parameters P, k, β, α, N_s **Output**: Final subset of bands $\mathbf{B}_f$

Using k-means, cluster d bands of $\mathbf{X}$ into P clusters $\{\mathcal{S}_i\}_{i=1}^{P}$.

Partition the image $\mathbf{X}$ into N_s homogeneous and disjoint superpixels using SLIC algorithm.

Compute the label matrix $\mathbf{H} \in \mathbb{R}^{c \times m}$ of subset $\mathbf{X}$ using (3);

Find the cluster representatives $\mathbf{B}_i$ of each cluster S_i using the following approach:

for $i = 1{:}P$ **do**

  Let $\mathbf{X}_i \in \mathbb{R}^{d_i \times m}$ be the image subset made of d_i bands from cluster $\mathbf{S}_i$;

  Find the weight matrix $\mathbf{A}_i \in \mathbb{R}^{m \times m}$ for $\mathbf{X}_i$ using the superpixel-based graph Laplacian embedding discussed in (5). Using $\mathbf{A}_i$, find $\mathbf{L}_i$;

  Find the largest k principal components of $(\mathbf{X}_i{}^{\mathrm{T}}\mathbf{X}_i + \alpha\mathbf{H}^{\mathrm{T}}\mathbf{H} - \beta\mathbf{L}_i)$

  Compute the component loadings of all bands in cluster $\mathbf{S}_i$ for the chosen k principal components using (16);

  Select the band with highest component loadings as the cluster representative $\mathbf{B}_i$.

end for

$\mathbf{B}_f = \bigcup_{i=1}^{P} \mathbf{B}_i, \forall i = 1, \ldots, P.$

IV. Results and Discussion

To prove the effectiveness of the proposed method, a series of experiments are conducted. Three publicly available datasets have been used in the experiments. The datasets and the experimental design are discussed in Sections IV-A and IV-B. The sensitivity of the algorithm toward various parameters such as the number of clusters, size of superpixels, and weighing parameters is investigated in Section IV-C. This is followed by an objective comparison of the proposed approach with several widely used, state-of-the-art BS methods as well as some more recent methods in Section IV-D. Finally, the computational performance of all the methods is discussed and compared in Section IV-F where the computational complexity of the proposed method is compared with the existing methods.

A. Datasets Used

Three well-known and publicly shared datasets of diverse nature have been used in this study. The first scene, Pavia University [59], is acquired by the reflective optics system imaging spectrometer (ROSIS) during a flight campaign over Pavia, Northern Italy. The scene has 1.3-m spatial resolution, 610 × 610 samples, and 115 bands. Some of the samples have no information and have been discarded (as black pixels) from the image. Similarly, many noisy bands have been removed reducing the number of remaining bands to 103. The ground truth has nine classes. Pavia scenes were provided by Prof. Paolo Gamba from the Telecommunications and Remote Sensing Laboratory, Pavia University, Pavia, Italy. Salinas dataset [59] is captured using the 224-band, high spatial resolution (3.7-m) AVIRIS sensor over Salinas Valley, California and contains 512 × 217 samples. It contains 16 classes such as

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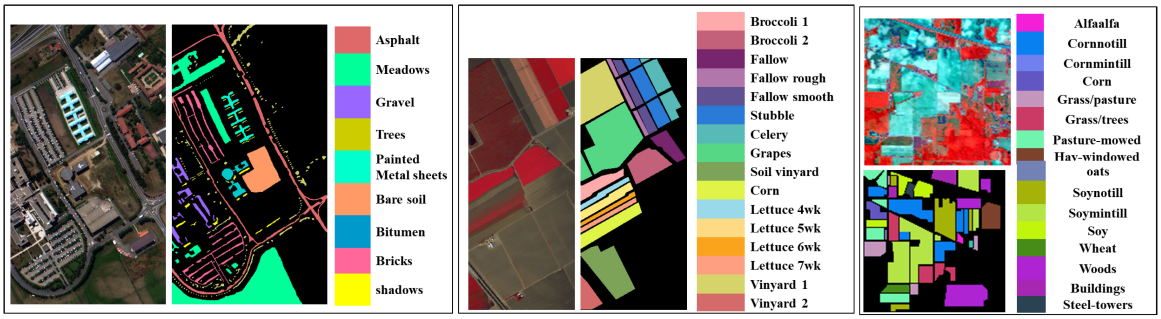


Fig. 2. Hyperspectral datasets used in the experiment (shown in FCC) along with their ground truth maps. (a) Pavia University ($R = 53$, $G = 31$, and $B = 8$). (b) Salinas ($R = 47$, $G = 27$, and $B = 13$). (c) Indian Pines ($R = 57$, $G = 27$, and $B = 17$).

TABLE I
HYPER-PARAMETER SETTINGS FOR DIFFERENT BS METHODS

Method	Hyper-parameters	Indian pines	Pavia	Salinas
ASPS	Size of image blocks, B_{im}	B_{im}=3	B_{im}=10	B_{im}=10
BS-Nets	Regularization coefficient λ Learning rate η Maximum iterations max_i	λ =1e-2 η =2e-3 max_i =100	λ =1e-2 η =2e-3 max_i =100	λ =1e-2 η =2e-3 max_i =100
LLRSC	Regularization parameters: λ, α	λ=1e-8 α=0.001	λ=1e-11 α=0.001	λ=1e-10 α=0.001
TMI-CSA	Mutation probability factor n_m Clonal scale factor n_c	n_m = 7000 n_c = 2500	n_m = 7000 n_c = 2500	n_m = 7000 n_c = 2500
EGCSR	Regularization parameter λ Number of nearest neighbours k_n	λ=1e4 k_n=5	λ=1e4 k_n=5	λ=1e4 k_n=5
JAYA	Population P Number of iterations N_i Upper bounds U_b Lower bounds L_b	P = 10 N_i =50 U_b =220 L_b = 1	P = 10 N_i =50 U_b=103 L_b= 1	P = 10 N_i =50 U_b=224 L_b=1
Proposed	Number of clusters P Number of principal components k Number of superpixels N_s Regularization coefficient α, β	P=10 k=3 N_s=500 $\alpha = \beta$ = 0.5	P=10 k=3 N_s=1000 $\alpha = \beta$ = 0.5	P=10 k=3 N_s=500 $\alpha = \beta$ = 0.5

vegetables, bare soils, and vineyard fields. However, 20 water absorption bands have been discarded during preprocessing and the remaining bands were used. Indian Pines dataset [59] is collected by AVIRIS sensor over the Indian Pines test site in North-western Indiana and consists of 145 × 145 pixels and 224 spectral reflectance bands. The Indian Pines scene contains agricultural, forest and other natural perennial vegetation classes. The ground truth is divided into sixteen classes. The number of bands was reduced to 200 by removing bands covering the region of water absorption. The false color composite (FCC) of the scenes are shown in Fig. 2(a)–(c).

B. Experimental Design

For segmenting the images into superpixels, the SLIC algorithm is used, with compactness parameter $\tau = 25$ for all the datasets and number of superpixels $N_s = 1000$ for Pavia university dataset and $N_s = 500$ for both Salinas and Indian Pines datasets.

Next, an SVM classifier using radial basis kernel function is used for the classification in this study. For the comparison, fivefold cross validation [60] method is used where the dataset is randomly partitioned into five subsets of equal size. In each run, one subset of the dataset is used as training data while the rest is used as test data. The accuracy is taken as the average accuracy (AA) across all the subsets. Three performance indicators, namely, overall accuracy (OA), AA, and kappa coefficient are used for the assessment of classification results.

The proposed algorithm is compared with the seven latest and state-of-the-art methods. These methods are Laplacian-regularized low-rank subspace clustering (LLRSC) [61], band selection networks (BS-Nets) [62], BS via adaptive subspace partition strategy (ASPS) [63], trivariate mutual information clonal selection algorithm (TMICSA) [32], EGCSR [26] and spectral clustering and spatial Frobenius norm-based Jaya Optimization (JAYA) 45]. The last two methods fall into the category of spectral–spatial BS methods. The different hyperparameter settings for the proposed algorithm as well as the other BS algorithms compared in this study are listed in Table I.

Another important part of the experimental study is the investigation of the sensitivity of the algorithm toward different

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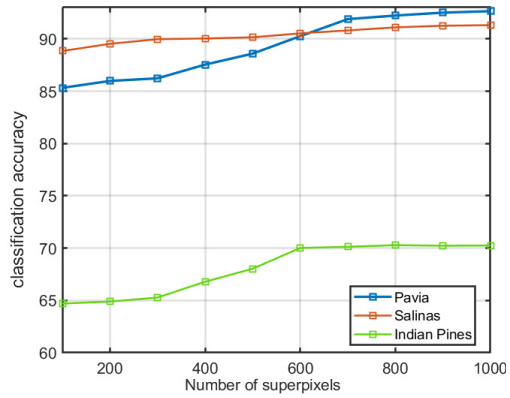


Fig. 3. Influence of number of superpixels N_s on the classification accuracy. The decline in classification accuracy is greater for heterogeneous urban scenes such as Pavia University.

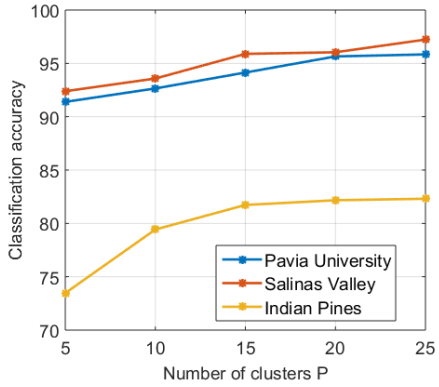


Fig. 4. Influence of number of clusters P on the classification accuracy. An increase in cluster size results in an improvement in classification.

tunable parameters. Some of the important parameters in the methodology are number of clusters P, size of superpixels N_s, weighing parameters α, β, etc. Each parameter is investigated by fixing the other parameters and observing the variation in the classification accuracy over a predefined range of the said parameter.

C. Parameter Analysis

An important aspect of the experiment is to investigate the respective roles of different parameters used in the experiment. Once the sensitivity of the algorithm toward these parameters is understood, they can be appropriately tuned to achieve optimal results. Some of the parameters evaluated here are: number of clusters P, number of superpixels N_s (segmentation scale) and weighing parameters α, β. Each parameter is investigated by fixing the other parameters and observing the variation in the classification accuracy over a predefined range of values of the said parameter.

For the sensitivity analysis, the following parameter settings are used: The different values of cluster number P was taken as $P = \{5, 10, 15, 20, 25\}$. To study the influence of superpixel/neighborhood size, the image was segmented into different number of superpixels within the following range: $N_s = \{100, 200, \ldots, 1000\}$. The different values of α and β, the scalars controlling the relative importance of each term in the objective function are taken as $\alpha = \{0.01, 0.1, 1, 10, 100, 1000\}$ and $\beta = \{0.01, 0.1, 1, 10, 100, 1000\}$.

1) Influence of N_s: The number of superpixels N_s dictate the segmentation scale which can influence the proposed BS algorithm, which in turn, affects the classification accuracy. N_s also indicates the size of neighborhood used for computing the weight matrix A. Fig. 3 shows the influence of N_s on the classification accuracy. The classification accuracies for all the three datasets are compared here. For this experiment, the number of clusters $P = 5$ is used. A larger scale, or smaller values of N_S leads to multiple classes or landscapes within the same superpixel. Thus, spatial relationship between a sample and its neighbors within the same superpixel is not properly represented due to incorrect computation of the similarity matrix A. Due to this, lower values of classification are observed for both datasets at smaller values of N_s.

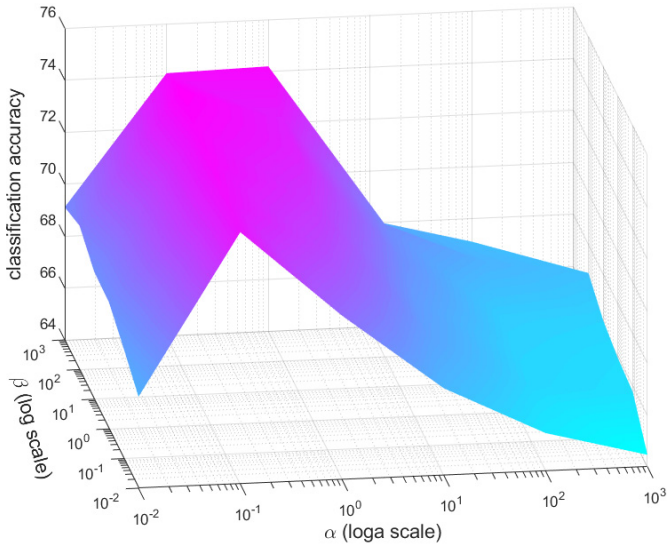


Fig. 5. Change in classification accuracy for different range of α and β values.

The classification accuracies become consistent beyond a given value of N_s. For urban scenes such as Pavia University consisting of spectrally heterogeneous objects such as buildings, roads, trees, water bodies, etc., higher values of N_S are required for the accurate computation of A. Thus, the classification of Pavia University achieves consistency at a higher value of N_s ($\geq$800) in comparison to Salinas Valley dataset ($\geq$500) or Indian pines ($\geq$600). Based on this, the values of N_s for Pavia University, Salinas Valley, and Indian Pines datasets for the rest of the experiments are set as 1000, 500, and 500, respectively.

2) Influence of P: The performance of the proposed algorithm is evaluated for a different number of clusters P, while keeping the other hyperparameter constant (see Fig. 4). For all three datasets, increasing the number of clusters results in an improvement in the classification accuracy. The change in accuracy is more significant for Indian Pines than the other two datasets. Indian pines is a challenging dataset consisting of

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TABLE II
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS FOR PAVIA UNIVERSITY DATASET

Class	LLSRC	BS-Nets	ASPS	TMI-CSA	EGCSR	JAYA	Proposed	All
Asphalt	92.61	90.26	93.25	92.65	88.34	90.53	94.72	91.89
Meadows	93.74	94.29	85.84	97.12	95.64	96.24	97.05	97.46
Gravel	64.13	72.65	69.58	81.6	70.35	73.99	88.11	71.77
Trees	86.07	88.26	92.56	95.62	91.44	91.28	92.71	90.48
Metal Sheets	99.75	98.26	83.25	94.25	99.75	99.65	100	99.45
Bare soil	64.74	85.69	86.56	92.51	76.62	81.54	87.61	83.17
Bitumen	79.20	78.26	83.54	87.99	78.95	80.44	85.96	83.61
Bricks	83.44	84.75	80.24	88.56	84.58	83.64	91.24	83.59
Shadows	100.00	98.56	89.26	97.43	100	99.88	99.65	100.00
OA	92.56	88.40	83.67	93.4	89.48	90.79	95.84	90.46
AA	87.14	87.88	84.90	91.97	87.29	88.57	93.01	89.05
kappa	0.84	0.86	0.89	0.92	0.86	0.87	0.92	0.88

TABLE III
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS FOR SALINAS VALLEY DATASET

Class	LLSRC	BS-Nets	ASPS	TMI-CSA	EGCSR	JAYA	Proposed	All
Broccoli 1	97.04	96.26	90.23	98.49	97.26	98.64	99.83	99.50
Broccoli 2	97.51	98.78	85.67	98.86	98.65	97.85	99.91	99.76
Fallow	96.27	98.76	94.29	97.23	97.25	100	98.67	99.66
Fallow(rough)	97.65	97.87	90.68	97.9	97.21	98.45	99.76	98.96
Fallow(smooth)	95.98	97.65	91.73	97.59	96.15	97.23	99.38	98.30
Stubble	96.83	98.72	93.65	97.32	97.26	96.67	99.75	99.94
Celery	75.32	98.19	92.34	96.47	99.87	98.45	99.72	99.47
Grapes	95.09	87.26	88.89	85.06	86.16	87.85	88.11	85.00
Soil vinyard	93.03	88.51	80.39	98.61	97.16	98.97	99.84	99.89
Corn	95.31	98.42	92.27	97.01	96.15	99.51	96.54	94.64
Lettuce (4wk)	95.48	95.65	92.35	95.81	96.49	97.29	99.06	99.17
Lettuce (5wk)	90.55	92.89	92.61	98.25	99.73	99.99	99.83	99.60
Lettuce (6wk)	95.16	99.54	91.99	100	97.98	98.18	99.64	98.42
Lettuce (7wk)	90.66	95.68	93.21	95.65	95.18	97.96	97.20	94.39
Vinyard 1	75.41	70.86	75.26	73.91	65.89	72.11	75.55	73.41
Vinyard 2	95.33	96.45	89.27	96.56	96.26	99.5	98.23	99.57
OA	90.25	95.40	87.45	93.27	91.56	93.97	96.13	93.77
AA	92.33	94.46	89.68	95.29	94.66	96.16	97.93	96.23
kappa	0.92	0.90	0.88	0.91	0.90	0.93	0.93	0.92

many similar classes. Many low-variance bands which can find subtle differences between such classes have low loadings with the top principal components of each cluster and are rejected in the BS process. By increasing the number of clusters, these bands have a higher chance of selection, thus, improving the classification accuracy.

3) Influence of α and β: Next, the classification performance of the algorithm is evaluated for different values of weighing coefficients α and β, while keeping the other hyperparameters constant (see Fig. 5). While α governs the contribution of class information, β is responsible for the induction of local spatial information in the objective function of PCA. The different values of α and β, the scalars controlling the relative importance of each term in the objective function is taken as $\alpha = \{0.01, 0.1, 1, 10, 100, 1000\}$ and $\beta = \{0.01, 0.1, 1, 10, 100, 1000\}$. From Fig. 5, it can be seen that as the values of α and β increase, the classification accuracy increase and then start decreasing beyond a particular value. In this case (Indian Pines dataset), the values of $\alpha = 0.1$ and $\beta = 100$ are giving the best classification performance. Fig. 5 also proves that α may have a greater role than β in the BS procedure. Thus, the proposed algorithm is relatively robust to β and can obtain a relatively stable performance over a larger range of this parameter. The above observation can also corroborate an assumption that since the selected bands are used for the classification of hyperspectral images, it is more important for these bands to have greater discriminative information than spatial information.

D. Comparison With Existing Approaches

The proposed methodology is compared with seven state-of-the-art methods (listed in Section IV-B). For a fair comparison, it is ensured that the number of bands is kept the same for all the methods. The percentage of training samples is also kept the same for all methods.

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TABLE IV
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS FOR INDIAN PINES DATASET

Class	LLRSC	BS-Nets	ASPS	TMI-CSA	EGCSR	JAYA	Proposed	All
Alfaalfa	68.28	58.26	49.75	52.1	31.820	43.90	71.43	8.89
Corn(no till)	70.54	77.84	63.26	78.53	54.97	64.67	73.83	51.70
Corn(min. till)	74.62	58.67	64.25	71.89	47.59	62.12	71.77	50.68
Corn	60.89	54.13	49.88	63.89	37.78	42.25	64.59	26.09
Grass/pasture	83.65	84.05	75.91	94.55	79.08	87.13	97.22	54.80
Grass/trees	90.54	90.26	75.93	93.32	93.22	91.17	97.25	94.77
Pasture(mowed)	86.15	52.69	56.27	51.85	55.56	56	100	48.15
Hay(windowed)	92.62	93.28	59.61	97.18	96.04	96.05	98.61	99.57
Oats	56.78	25.86	84.21	10.93	10.53	16.67	41.26	15.79
Soy(no till)	74.12	68.54	42.13	70.61	67.50	72.80	78.08	72.43
Soy(min. till)	74.36	73.51	65.99	82.59	73.58	78.32	87.50	72.70
Soy	73.54	66.21	86.56	73.54	36.59	51.87	79.78	36
Wheat	89.72	90.68	86.24	93.52	88.72	98.37	100	92.46
Woods	88.39	86.24	85.21	98.26	95.34	94.46	93.16	95.60
Buidings	69.28	49.26	52.63	40.55	36.51	34.29	55.17	22.19
Steel towers	87.36	83.54	75.65	82.59	97.73	95.24	92.86	90
OA	75.95	70.46	69.51	82.13	69.49	74.93	84.66	67.75
AA	77.55	69.56	67.09	72.24	62.66	67.83	78.78	58.24
kappa	0.78	0.72	0.73	0.78	0.65	0.72	0.81	0.63

The complete comparative analysis is a three-part experiment. First, an objective comparison of the proposed method with other BS methods is done. The accuracies of all the methods are expressed in terms of OA, AA, and kappa values. The next experiment involves the comparison of classification accuracies of all the methods for different training sizes. For this part, the size of the band subset was fixed at 30. The section is concluded by comparing the classification results of the methods for different sizes of band subset. For this part of the experiment, 10 percent of the labeled samples were used for training and the rest were utilized for testing.

The classification results for the three datasets are shown in Tables II–IV. The selected bands for all methods for each data can be seen as Figs. S1–S3 in the supplemental material. As can be seen from the three tables, the proposed methodology has given highest values of OA and AA for all three datasets. It can be observed from the tables that all the methods have shown better classification performance for Pavia and Salinas dataset, in comparison to Indian Pines dataset, which is known to be a challenging dataset. Indian pines dataset consists of vegetative classes of different varieties but with many similar spectral reflectances with the exception at few bands. Thus, it is important to retain bands with strong discriminative capabilities to differentiate between these similar classes.

For the Pavia dataset (see Table II), all the BS methods show the highest accuracies in highly reflective classes such as metal sheets and shadow class with extremely low reflectance since these classes are spectral distinctive from the rest of the classes. Classes such as gravel, bare soil, bitumen, and bricks have lower classification accuracies in all the methods as these classes are spectrally confusing with each other. However, for most of these classes, the proposed method has demonstrated better accuracies than the rest of the methods. In this case, the proposed method, along with JAYA have shown overall good performance due to their spectral–spatial BS criterion. Among the spectral-based BS approach, TMI-CSA has shown very good results.

The classification accuracies are highest for Salinas dataset out of all the three datasets. Almost all the BS methods have given consistently good performances for all classes of this dataset (see Table III). Noticeably, out of 16 classes, the proposed method has given highest accuracies for seven classes in total and over 99 percent accuracy in 10 out of 16 classes. The proposed method has the highest OA, AA, and kappa for this dataset followed by BS-Nets and JAYA.

For the Indian pines dataset, both the proposed method as well as TMI-CSA have given very good performances followed by LLRSC and JAYA. The proposed method has the highest OA, AA, and kappa values. Although, TMI-CSA, LLRSC, and JAYA also show good overall performances, it is the proposed method which shows the highest class-wise accuracies in most of the cases. In the case of datasets such as Indian Pines consisting of several classes with very similar spectral behavior, the use of only spectral information is not sufficient to discriminate between these classes.

By incorporating a selection criterion based on the spatial relationship between the neighboring pixels, the proposed algorithm has shown an improvement in the classification performance, as evident from Table III. In this manner, misclassification of anomalous or spectrally uncertain samples can be controlled, resulting in smoother and less noisier classification. Second, the use of labeled information of the training samples in the modified objective function of PCA has helped in enhancing the discriminative capability of the BS model. Finally, to ensure that the low variance bands with subtle yet vital discriminative information get equal opportunity in the BS process, the bands are first divided into different clusters of similar bands, and bands are selected from each group using the loadings of the basis function for each group.

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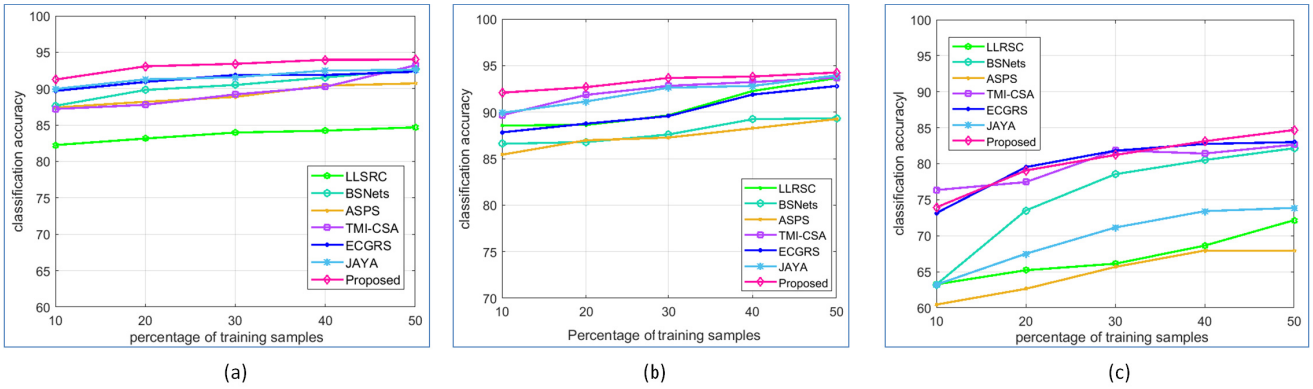


Fig. 6. Classification accuracy of different methods for training size ranging between $(10, 20, \ldots, 50)$ for (a) Salinas Valley, (b) Pavia University, and (c) Indian Pines dataset.

Finally, from Tables II–IV, the classification performance using the selected bands outweighs the classification performance using all the available bands. Thus, this means that many of the bands in the dataset are not only unnecessary but also detrimental to the classification and their removal using a suitable DR method is necessary to enhance the classification accuracy.

Next, the classification accuracies are compared under different training sizes ranging $(10, 20, \ldots, 50)$ percentage, while keeping the number of bands fixed. The results are shown for all datasets in Fig. 6. For all three datasets, the classification performance expectedly improves with the increase in training samples. Most of the methods reach consistency after reaching 25–30 percent training size in Pavia and Salinas dataset. Indian Pines, being a challenging dataset, requires more labeled samples. In the case of purely spectral-based BS methods, TMI-CSA and BSNets show the best results. Out of all three spectral–spatial BS, the proposed approach shows the best performance followed by JAYA. The performance of JAYA, however, falls down for the Indian Pines dataset. Overall, the superior performance of the proposed approach shows that the inclusion of class information as well as neighborhood relationship has led to a significant improvement in the classification results.

The last part of this experiment is the comparison of the BS methods under different band subset sizes ranging from 10 to 50, with keeping the training size as 10 percent (see Fig. 7). For the proposed method, the number of bands was varied by adjusting the cluster size. As it is evident, the method showed consistent and stable classification performance after the number of bands reaches 35. The proposed method, along with JAYA and TMI-CSA show good classification performance even for smaller band size in comparison to the other methods.

E. Computational Time and Complexity Analysis

Here, the impact of different parameters on the computational complexity of the proposed algorithm is discussed followed by a comparison son of the running times of all methods (see Table V). The methods are programmed in MATLAB 2015a and run on an Intel Xeon E5-2630 CPU machine at 2.20 GHz with 16 GB of RAM. However, both

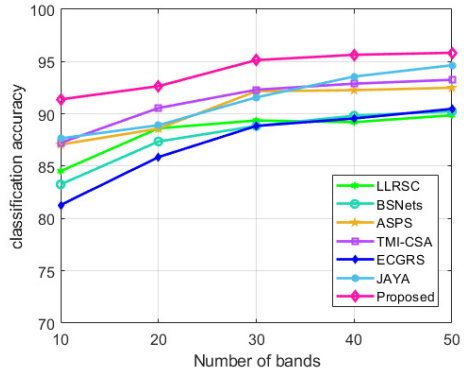


Fig. 7. Classification accuracy of different methods for number of bands ranging between $(10, 20, \ldots, 50)$.

BS-Net as well as EGCSR are programmed with Python 3.5 (code available online).

Assuming that d is the number of bands, m is the number of training samples, k and P are the number of principal components and number of clusters, respectively, and I is the number of iterations, the complexity of the proposed method can be explained as follows: The computational complexity of k-means clustering is $O(dIP)$ and the complexity of SLIC algorithm is $O(m)$. Although, the complexity of computing the similarity matrix A of size $m * m$ is $O(m^2)$, assuming that average size of a superpixel is m/N_s, the complexity reduces to $O(m^2/N_s^2)$. Next, the complexity of eigenvalue decomposition is $O(m^3)$, respectively. Thus, the complexity associated with eigenvalue decomposition of (13) becomes $O(m^3) + O(m^2/N_s)$ per cluster and thus, $O(Pm^3) + O(P(m^2/N_s))$ for all P clusters. The complexity becomes approximately $O(dIP) + O(m) + O(Pm^3) + O(P(m^2/N_s))$. Assuming that $m \gg N_s \gg d > P$, the complexity can be roughly approximated as $O(Pm^3)$.

The computational complexities of some of the compared methods, as explained by their authors, are discussed as follows: For ASPS, computational complexity is given as $O(d^2m + dm)$ [63]. Similarly, after understanding the algorithm of LLRSC [61], it can be assumed that its computational complexity is roughly similar to that of ISSC [64], which is, $O(d^2m + d^2 + Pmt1)$, $t1$ is the number of iterations defined for the algorithm. As mentioned in [62], BS-Nets is

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TABLE V
COMPUTATIONAL TIME (IN SECONDS) OF DIFFERENT BS METHODS

Method	Indian Pines	Pavia University	Salinas
ASPS	1.97	4.56	5.69
BS-Nets	380	596	601
LLRSC	15.35	12.73	14.29
TMI-CSA	96.79	47.66	140.86
EGCSR	5.25	20.16	23.96
JAYA	39.97	70.52	49.29
Proposed	221	268	327

computationally more complex than algebraic methods like ISSC but can be easily accelerated with GPU. Since EGCSR has a closed-form solution, it shows much less computational burden than BS-Net [26]. Thus, it can be deduced that the computational complexities of the proposed method, BS-Nets and TMI-CMA are higher than LLRSC and ASPS. In case of the proposed method, the complexity increases for higher number of samples.

From Table V, it can be seen that the running times of the proposed method, JAYA, TMI-CSA, and BS-Nets are significantly higher in comparison to the remaining methods which can also be corroborated from their higher computational complexities from the rest of the methods. The number of trainable parameters and training samples required in BS-Nets is quite high, increasing the computational time significantly. However, as quoted by the authors, the BS-Nets algorithm can also be accelerated with GPUs [62].

V. CONCLUSION

In this article, a supervised BS method based on component loadings obtained using spectral–spatial PCA is proposed. The band importance is determined using the component loadings of the PCs estimated from a unified objective function having three terms: data fidelity, classification error term and spatial prior. A novel superpixel-based graph model is introduced to incorporate the spatial information into the PCA objective function, which also minimizes the search space and time. Using three hyperspectral datasets, comprehensive experiments are conducted to compare the performance of the proposed methodology with several state-of-the-art and newer BS methods (both spectral only as well as spectral–spatial type). The methodology has shown better and more stable performance for different training sizes and number of bands. The role of different parameters governing the selection of bands is also investigated. Finally, the computational complexities and running times of the proposed method with the other methods were compared.

APPENDIX I
PROOF OF SOLUTION TO EQUATION (4)

From (6)

$$\begin{aligned}
&\min_{\mathbf{Q}} \frac{\sum_{i,j=1}^{m} \left\|\mathbf{q}_i - \mathbf{q}_j\right\|^2 \mathbf{A}(i,j)}{2} \\
&= \frac{1}{2}\min_{\mathbf{Q}} \left\{ \sum_{i=1}^{m} \mathbf{q}_i \cdot \mathbf{q}_i \sum_{j=1} \mathbf{A}(i,j) \right. \\
&\qquad \left. + \sum_{j=1}^{m} \mathbf{q}_j \cdot \mathbf{q}_j \sum_{i=1} \mathbf{A}(i,j) - 2\sum_{i,j=1} \mathbf{q}_i \cdot \mathbf{q}_j \mathbf{A}(i,j) \right\} \\
&= \frac{1}{2}\min_{\mathbf{Q}} \left\{ \sum_{i=1}^{m} \mathbf{q}_i \cdot \mathbf{q}_i \mathbf{D}(i,i) \right. \\
&\qquad \left. + \sum_{j=1}^{m} \mathbf{q}_j \cdot \mathbf{q}_j \mathbf{D}(j,j) - 2\sum_{i,j=1} \mathbf{q}_i \cdot \mathbf{q}_j \mathbf{A}(i,j) \right\} \\
&= \frac{1}{2}\min_{\mathbf{Q}} \left\{ 2\sum_{i,j=1} \mathbf{q}_i \cdot \mathbf{q}_j \mathbf{L}(i,j) \right\} \\
&= \min_{\mathbf{Q}} \left\{ \mathrm{Tr}\left(\mathbf{Q}^{\mathrm{T}}\mathbf{L}\mathbf{Q}\right) \right\}
\end{aligned}$$

where $\mathbf{L} = \mathbf{D} - \mathbf{A}$ is the Laplacian matrix, $\mathbf{D}$ is the degree matrix and tr$(\cdot)$ is the trace of the matrix. To eliminate the trivial solution of $\mathbf{q}_i = 0, \forall i = 0, 1, 2, \ldots, m$, $\mathbf{Q}^{\mathrm{T}}\mathbf{Q} = I$ is added as a constraint to the minimization problem.

APPENDIX II
PROOF OF EQUATION (13)

From (12)

$$\begin{aligned}
\mathcal{L}_1(\mathbf{Q}) &= \left\|\mathbf{X} - \mathbf{X}\mathbf{Q}\mathbf{Q}^{\mathbf{T}}\right\|_F^2 + \alpha\left\|\mathbf{H} - \mathbf{H}\mathbf{Q}\mathbf{Q}^{\mathbf{T}}\right\|_F^2 + \beta\,\mathbf{Q}^{\mathrm{T}}\mathbf{L}\mathbf{Q} \\
&= \mathrm{Tr}\left(\left(\mathbf{X} - \mathbf{X}\mathbf{Q}\mathbf{Q}^{\mathbf{T}}\right)^{\mathrm{T}}\left(\mathbf{X} - \mathbf{X}\mathbf{Q}\mathbf{Q}^{\mathbf{T}}\right)\right) \\
&\quad + \alpha\,\mathrm{Tr}\left(\left(\mathbf{H} - \mathbf{H}\mathbf{Q}\mathbf{Q}^{\mathbf{T}}\right)^{\mathrm{T}}\left(\mathbf{H} - \mathbf{H}\mathbf{Q}\mathbf{Q}^{\mathbf{T}}\right)\right) + \beta\,\mathrm{Tr}\left(\mathbf{Q}^{\mathrm{T}}\mathbf{L}\mathbf{Q}\right) \\
&= -\mathrm{Tr}\left(\mathbf{X}^{\mathrm{T}}\mathbf{X}\mathbf{Q}\mathbf{Q}^{\mathrm{T}}\right) + \mathrm{Tr}\left(\mathbf{X}^{\mathrm{T}}\mathbf{X}\right) - \mathrm{Tr}\left(\mathbf{H}^{\mathrm{T}}\mathbf{H}\mathbf{Q}\mathbf{Q}^{\mathrm{T}}\right) \\
&\quad + \mathrm{Tr}\left(\mathbf{H}^{\mathrm{T}}\mathbf{H}\right) + \beta\,\mathrm{Tr}\left(\mathbf{Q}^{\mathrm{T}}\mathbf{L}\mathbf{Q}\right) \\
&= -\mathrm{Tr}\left(\mathbf{Q}^{\mathrm{T}}\mathbf{X}^{\mathrm{T}}\mathbf{X}\mathbf{Q}\right) - \mathrm{Tr}\left(\mathbf{Q}^{\mathrm{T}}\mathbf{H}^{\mathrm{T}}\mathbf{H}\mathbf{Q}\right) + \mathrm{Tr}\left(\mathbf{X}^{\mathrm{T}}\mathbf{X}\right) \\
&\quad + \mathrm{Tr}\left(\mathbf{H}^{\mathrm{T}}\mathbf{H}\right) + \beta\,\mathrm{Tr}\left(\mathbf{Q}^{\mathrm{T}}\mathbf{L}\mathbf{Q}\right) \\
&= -\mathrm{Tr}\left(\mathbf{Q}^{\mathrm{T}}\left(\mathbf{X}^{\mathrm{T}}\mathbf{X} - \alpha\mathbf{H}^{\mathrm{T}}\mathbf{H} + \beta\mathbf{L}\right)\mathbf{Q}\right) + \mathrm{Tr}\left(\mathbf{X}^{\mathrm{T}}\mathbf{X}\right) \\
&\quad + \mathrm{Tr}\left(\mathbf{H}^{\mathrm{T}}\mathbf{H}\right).
\end{aligned}$$

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